



Modelling Financial Markets

Seminar 4: Portfolio Optimisation — Hands-on



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Acronyms used in this material (1/2)

- ▣ AAPL: Apple Inc. (Nasdaq ticker)
- ▣ AI: Artificial Intelligence
- ▣ AMZN: Amazon.com, Inc. (Nasdaq ticker)
- ▣ BAC: Bank of America Corporation (NYSE ticker)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BET-TR: BET Total Return (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CSCO: Cisco Systems, Inc. (Nasdaq ticker)
- ▣ ERC: Equal Risk Contribution portfolio
- ▣ ETF: Exchange-Traded Fund
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GLS: Generalised Least Squares
- ▣ GME: GameStop Corp. (NYSE ticker)
- ▣ GMV: Global Minimum-Variance portfolio
- ▣ GMV-LW: Global Minimum-Variance portfolio with Ledoit–Wolf covariance
- ▣ GOOGL: Alphabet Inc., Class A shares (Nasdaq ticker)
- ▣ GS: The Goldman Sachs Group, Inc. (NYSE ticker)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HRP: Hierarchical Risk Parity
- ▣ HYG: iShares iBoxx High Yield Corporate Bond ETF
- ▣ IEF: iShares 7–10 Year Treasury Bond ETF
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ LQD: iShares iBoxx Investment Grade Corporate Bond ETF
- ▣ MS: Morgan Stanley (NYSE ticker)
- ▣ MSFT: Microsoft Corporation (Nasdaq ticker)

Acronyms used in this material (2/2)

- ▣ MSTR: **Strategy Inc., formerly MicroStrategy (Nasdaq ticker)**
- ▣ MV: **Mean–Variance tangency portfolio estimated from sample moments**
- ▣ MV-LO: **Mean–Variance tangency portfolio, Long-Only**
- ▣ NVDA: **NVIDIA Corporation (Nasdaq ticker)**
- ▣ OLS: **Ordinary Least Squares**
- ▣ OOS: **Out-Of-Sample**
- ▣ US: **United States**
- ▣ WFC: **Wells Fargo & Company (NYSE ticker)**

Outline

Introduction

Key Definitions

Part A: Derivations

Part B: Backtests and Inference

Part C: Open Question

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive GMV, ERC and Black–Litterman portfolios and the bias of the in-sample Sharpe ratio
- ▣ Run an out-of-sample backtest with a rolling window, turnover and costs
- ▣ Test a difference in Sharpe ratios with HAC errors and a block bootstrap
- ▣ Judge whether an optimised portfolio really beats 1/N

- ▣ Background: Lecture 4; textbook: [Franke, Härdle & Hafner \(2019\)](#), Part I
- ▣ Seminar: Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Structure

- ▣ **Part A:** nine derivations (A1–A9)
- ▣ **Part B:** nine computational exercises, US and Bitcoin data (B1–B9)
- ▣ **Part C:** one open question on BVB portfolios, a starting point for the team project (C1), a critique of an AI answer (C2), and a replication of a published study on our data (C3)

How to Work in This Seminar

- ▣ **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- ▣ Every Part B exercise ends with an **interpretation question**
 - ▶ It cannot be answered by reading a number: ask what would change the conclusion
- ▣ Out of sample means: the weights of month t use only data up to month $t - 1$
- ▣ Report every Sharpe ratio difference with a p -value or an interval

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (GMV for N assets), A3 (ERC with two assets), A5 (Black–Litterman as GLS), A7 (variance of a Sharpe difference)
- Part B: B1 (1/ N versus Markowitz, sectors), B2 (shrinkage and GMV risk), B3 (ERC and HRP, sectors)

Proposed: you solve them

- Part A: A2 (model: A1); A4 (model: A3); A6 (model: A5); A8, A9 (model: A7)
- Part B: B4 (model: B2); B5 (model: B3); B6, B8 (model: B1); B7 (model: A5, A6); B9 (model: B1, A2)
- Part C: C1, open question (model: B1, B3); C2, critique of an AI answer (model: A7, B1); C3, replication of [Ledoit & Wolf \(2017\)](#) (model: B2, A7)

Setup: Python Environment and Data

-  **Open in Colab:** `chapter4_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `matplotlib`
- Series: US sector and multi-asset ETFs, Bitcoin, eight BVB blue chips, BET-TR (course repository, `data/market`)
- One-month risk-free rate: [Kenneth French Data Library](#); USD/RON reference rate: [BNR](#)
- ETFs and stocks: dividend-adjusted prices; indices and Bitcoin: closing values
- Joint analyses: join the *prices* on common days first, then take month-end returns

Key Definitions: Portfolios

- Mean–variance (Markowitz (1952)): tangency $w_{MV} \propto \Sigma^{-1}\mu$; GMV $w_{GMV} \propto \Sigma^{-1}1$; both normalised to sum to one
- Two assets: $w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$
- Risk contribution $RC_i = w_i(\Sigma w)_i/\sigma_p$; ERC (Maillard, Roncalli & Teïletche (2010)): equal RC_i
- HRP (López de Prado (2016)): correlation tree, then recursive bisection with inverse-variance weights
- Ledoit–Wolf (Ledoit & Wolf (2004)): $\hat{\Sigma} = \delta F + (1 - \delta)S$, constant-correlation target F
- Black–Litterman (Black & Litterman (1992)): prior $\pi = \delta_{RA}\Sigma w_b$, posterior μ_{BL} = precision-weighted mean of π and the views

Key Definitions: Evaluation and Inference

- ▣ Sharpe ratio $SR = \bar{r}^e / s(r^e)$, annualised $\times \sqrt{12}$; turnover $TO_t = \sum_i |w_{i,t} - w_{i,t-1}^+|$, $w^+ =$ drifted weights
- ▣ Net return: $r_t^{\text{net}} = r_t - c TO_t$, c the cost per unit of turnover as a decimal; a quote of 20 bp means $c = 0.002$ (1 bp = 0.01% = 10^{-4})
- ▣ i.i.d. test (Jobson & Korkie (1981); formula in Ledoit & Wolf (2008)), monthly SR_1, SR_2 , correlation ρ

$$\text{Var}(\widehat{SR}_1 - \widehat{SR}_2) \approx \frac{1}{T} \left[2 - 2\rho + \frac{1}{2} (SR_1^2 + SR_2^2 - 2SR_1SR_2\rho^2) \right]$$

- ▣ HAC test (Ledoit & Wolf (2008), Section 3.1): delta method with a Newey–West covariance of (r_1, r_2, r_1^2, r_2^2)
- ▣ Studentized circular block bootstrap (Ledoit & Wolf (2008), Section 3.2.2): pairs resampled in blocks, block length calibrated among $\{1, 2, 4, 6, 8, 10\}$, p -value from 4999 draws

Exercise A1: The GMV Portfolio for N Assets [Solved]

Task

- ▣ **Data and assumptions:** N risky assets with a known, positive definite covariance matrix Σ ; short sales allowed; weights sum to one. Numbers: SPY and TLT, monthly total returns August 2002 – July 2026 (data/market, $T = 288$), $\sigma_1 = 14.72\%$, $\sigma_2 = 13.49\%$, $\rho = -0.100$, means 11.73% and 4.32% a year
- ▣ (a) Write the Lagrangian of $\min_w w^\top \Sigma w$ subject to $w^\top \mathbf{1} = 1$ and derive $w_{\text{GMV}} = \Sigma^{-1} \mathbf{1} / A$, with $A = \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$
- ▣ (b) Show that the minimum variance equals $\sigma_{\text{GMV}}^2 = 1/A$
- ▣ (c) Invert the 2×2 matrix and show that for $N = 2$ the weight of asset 1 is $w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2) / (\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$
- ▣ (d) Compute A , w^* , σ_{GMV} and the mean of the GMV mix for SPY and TLT
- ▣ **Report:** the derivation, the four numbers and one sentence on why the GMV volatility is below both assets

Exercise A1: Solution (1/2)

Solution: derivation

- (a) $\mathcal{L} = \mathbf{w}^\top \Sigma \mathbf{w} - \lambda(\mathbf{w}^\top \mathbf{1} - 1)$; the first-order condition is $2\Sigma \mathbf{w} - \lambda \mathbf{1} = 0$, so $\mathbf{w} = \frac{\lambda}{2} \Sigma^{-1} \mathbf{1}$
- Imposing $\mathbf{1}^\top \mathbf{w} = 1$ gives $\frac{\lambda}{2} A = 1$, hence $\frac{\lambda}{2} = 1/A$ and $\mathbf{w}_{\text{GMV}} = \Sigma^{-1} \mathbf{1}/A$; the objective is strictly convex, so this is the unique minimum
- (b) $\sigma_{\text{GMV}}^2 = \mathbf{1}^\top \Sigma^{-1} \Sigma \Sigma^{-1} \mathbf{1}/A^2 = A/A^2 = 1/A$
- (c) With $\sigma_{12} = \rho\sigma_1\sigma_2$: $\Sigma^{-1} = \frac{1}{\det \Sigma} \begin{pmatrix} \sigma_2^2 & -\sigma_{12} \\ -\sigma_{12} & \sigma_1^2 \end{pmatrix}$, so $\Sigma^{-1} \mathbf{1} = \frac{1}{\det \Sigma} (\sigma_2^2 - \sigma_{12}, \sigma_1^2 - \sigma_{12})^\top$
- Dividing the first element by the sum of both gives $w^* = (\sigma_2^2 - \rho\sigma_1\sigma_2)/(\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2)$: the formula of the lecture recap is a special case

Exercise A1: Solution (2/2)

Solution: numbers

- ▣ Inputs: $\sigma_1^2 = 0.1472^2 = 0.02166$, $\sigma_2^2 = 0.1349^2 = 0.01820$, $\sigma_{12} = -0.100 \times 0.1472 \times 0.1349 = -0.00199$
- ▣ $\det \Sigma = 0.02166 \times 0.01820 - 0.00199^2 = 0.000390$; $\Sigma^{-1} \mathbf{1} = (0.02019, 0.02366)/0.000390 = (51.7, 60.6)$
- ▣ $A = 51.7 + 60.6 = 112.4$; $w^* = 51.7/112.4 = 0.461$ in SPY and 0.539 in TLT
- ▣ $\sigma_{\text{GMV}} = 1/\sqrt{112.4} = 9.43\%$; mean $0.461 \times 11.73\% + 0.539 \times 4.32\% = 7.73\%$
- ▣ Interpretation: with a negative correlation the two risks partly cancel, so the mix is 36% less volatile than SPY while keeping two-thirds of its mean

Exercise A2: Tangency Weights by Regression [Proposed]

Solution discussed in class

Task

- **Data and assumptions:** X is the $T \times N$ matrix of excess returns, rows $r_t^\top$; $\hat{\mu} = X^\top 1/T$ and $\hat{\Sigma} = X^\top X/T - \hat{\mu}\hat{\mu}^\top$ (maximum likelihood); $\hat{\Sigma}$ is invertible
- (a) Write the OLS estimator $\hat{b}$ of the regression $1 = b^\top r_t + u_t$ (no constant term) as a function of $\hat{\mu}$ and $\hat{\Sigma}$
- (b) Apply Sherman–Morrison, $(A + uu^\top)^{-1} = A^{-1} - A^{-1}uu^\top A^{-1}/(1 + u^\top A^{-1}u)$, and show that $\hat{b} = \hat{\Sigma}^{-1}\hat{\mu}/(1 + \hat{\theta}^2)$, $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1}\hat{\mu}$ (Britten-Jones (1999))
- (c) Show that $\hat{b}/1^\top \hat{b}$ equals the sample tangency weights when $1^\top \hat{\Sigma}^{-1}\hat{\mu} > 0$
- (d) State which hypothesis the OLS t -test of $b_i = 0$ tests, and write the restriction matrix R of the F -test of “tangency = $1/N$ ”
- **Report:** the proof in four lines and the two hypotheses; **Model:** A1

Exercise A3: ERC with Two Assets [Solved]

Task

- ▣ **Data and assumptions:** two assets, volatilities $\sigma_1, \sigma_2 > 0$, correlation $\rho \in (-1, 1)$, weights $w_1 + w_2 = 1$, long only; risk contribution $RC_i = w_i(\Sigma w)_i / \sigma_p$. Numbers: SPY ($\sigma_1 = 14.72\%$) and IEF ($\sigma_2 = 6.64\%$), $\rho = -0.100$, monthly August 2002 – July 2026
- ▣ (a) Write $RC_1\sigma_p$ and $RC_2\sigma_p$ explicitly for two assets
- ▣ (b) Prove that $RC_1 = RC_2$ if and only if $w_1\sigma_1 = w_2\sigma_2$, for any $\rho \in (-1, 1)$, so that ERC is inverse volatility
- ▣ (c) Compute the ERC weight of SPY and the volatility of the ERC portfolio; check that each asset carries half of the variance
- ▣ **Report:** the proof, w_1 , σ_{ERC} and the two risk shares

Exercise A3: Solution

Solution

- ▣ (a) $RC_1\sigma_p = w_1^2\sigma_1^2 + w_1w_2\rho\sigma_1\sigma_2$ and $RC_2\sigma_p = w_2^2\sigma_2^2 + w_1w_2\rho\sigma_1\sigma_2$
- ▣ (b) The cross terms are identical, so $RC_1 = RC_2 \iff w_1^2\sigma_1^2 = w_2^2\sigma_2^2 \iff w_1\sigma_1 = w_2\sigma_2$ (positive weights); ρ drops out
- ▣ (c) $1/\sigma_1 = 6.79$, $1/\sigma_2 = 15.06$: $w_1 = 6.79/(6.79 + 15.06) = 0.311$ in SPY, 0.689 in IEF; the numerical ERC solver gives the same 0.311
- ▣ $\sigma_p^2 = 0.311^2(0.02166) + 0.689^2(0.00441) + 2(0.311)(0.689)(-0.00098) = 0.00377$, $\sigma_{ERC} = 6.14\%$; risk shares 50% and 50%
- ▣ Interpretation: equal risk needs more than twice as much of the bond ETF, because it is less than half as volatile

Exercise A4: Who Carries the Risk in 60/40? [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** SPY and IEF as in A3, annualised $\sigma_1^2 = 0.02166$, $\sigma_2^2 = 0.00441$, $\sigma_{12} = -0.00098$; portfolio $w = (0.6, 0.4)$
- ▣ (a) Compute Σw , the volatility σ_p and the percentage risk shares $w_i(\Sigma w)_i / \sigma_p^2$ of the 60/40 portfolio
- ▣ (b) Compare σ_p with the ERC volatility of A3
- ▣ (c) Prove $\sum_i RC_i = \sigma_p$ with Euler's theorem for functions homogeneous of degree one
- ▣ (d) Prove that the ERC portfolio exists and is unique for Σ positive definite: show that $f(y) = \frac{1}{2}y^\top \Sigma y - \frac{1}{N} \sum_i \ln y_i$ is strictly convex on $y > 0$, tends to $+\infty$ at the boundary and at infinity, and that its first-order condition equalises the risk contributions
- ▣ **Report:** the numbers of (a)–(b) and the two proofs; **Model:** A3

Exercise A5: Black–Litterman as GLS [Solved]

Task

- **Data and assumptions:** prior $\pi = \mu + e_1$, views $q = P\mu + e_2$, $\text{Var}(e_1) = \tau\Sigma$, $\text{Var}(e_2) = \Omega$, e_1 and e_2 independent (Theil & Goldberger (1961)). Numbers: one asset, $\pi = 5\%$, $\sigma = 16\%$, $\tau = 0.05$, view $q = 8\%$ with standard deviation 2%
- (a) Stack the two equations as $y = Z\mu + e$ and derive the GLS estimator μ_{BL} and its covariance
- (b) Show that $P = I$ and $\Omega = \tau\Sigma$ give $\mu_{BL} = (\pi + q)/2$
- (c) For the one-asset numbers compute the two precisions, μ_{BL} , the weight of the view and the posterior standard deviation
- **Report:** the two formulas and the four numbers

Exercise A5: Solution

Solution

- (a) $y = (\pi^\top, q^\top)^\top$, $Z = (I, P^\top)^\top$, $V = \text{diag}(\tau\Sigma, \Omega)$; GLS: $\hat{\mu} = (Z^\top V^{-1}Z)^{-1}Z^\top V^{-1}y$
- Block multiplication: $\mu_{BL} = [(\tau\Sigma)^{-1} + P^\top \Omega^{-1}P]^{-1}[(\tau\Sigma)^{-1}\pi + P^\top \Omega^{-1}q]$, covariance $[(\tau\Sigma)^{-1} + P^\top \Omega^{-1}P]^{-1}$: the Black-Litterman formula of the lecture
- (b) $P = I$, $\Omega = \tau\Sigma$: $[2(\tau\Sigma)^{-1}]^{-1}(\tau\Sigma)^{-1}(\pi + q) = (\pi + q)/2$: equal precisions, equal weights
- (c) Prior variance $\tau\sigma^2 = 0.05 \times 0.0256 = 0.00128$, precision 781; view variance $0.02^2 = 0.0004$, precision 2500
- $\mu_{BL} = (781 \times 5\% + 2500 \times 8\%)/3281 = 7.29\%$; view weight $2500/3281 = 0.76$; posterior standard deviation $1/\sqrt{3281} = 1.75\%$
- Interpretation: the view is three times as precise as the prior, so it gets three-quarters of the weight; the posterior is more precise than either source

Exercise A6: The Sector View of the Lecture [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** nine sector ETFs, monthly excess returns August 2016 – July 2026, Σ annualised; benchmark $1/N$, $\delta_{RA} = 2.5$, $\tau = 0.05$; implied XLK minus XLU = $P\pi = 2.22\%$, $P\Sigma P^\top = 0.0477$
- ▣ (a) Write the prior distribution of the view portfolio $P\mu$ implied by $\mu \sim N(\pi, \tau\Sigma)$
- ▣ (b) With the view XLK minus XLU = 3% and $\Omega = \tau P\Sigma P^\top$ (He & Litterman (2002)), show that the posterior view is the simple average of prior and view, and compute it
- ▣ (c) Compute the new weights $w_{BL} = (\delta_{RA}\Sigma)^{-1}\mu_{BL}$ and say which weights change
- ▣ **Report:** the posterior view and the XLK and XLU weights; **Model:** A5

Exercise A7: The Variance of a Sharpe Difference [Solved]

Task

- ▣ **Data and assumptions:** two return series, i.i.d. bivariate Normal, means μ_i , variances σ_i^2 , correlation ρ , $SR_i = \mu_i/\sigma_i$ monthly. Numbers: sector backtest, $T = 271$ months (January 2004 – July 2026), monthly Sharpe GMV 0.2051, $1/N$ 0.1908, $\rho = 0.694$
- ▣ (a) Write the asymptotic covariance matrix of $\sqrt{T}(\hat{\mu}_1, \hat{\mu}_2, \hat{\sigma}_1^2, \hat{\sigma}_2^2)$ under Normality
- ▣ (b) Apply the delta method to $f = \mu_1/\sigma_1 - \mu_2/\sigma_2$ and derive
$$T \text{Var}(\widehat{SR}_1 - \widehat{SR}_2) \rightarrow 2 - 2\rho + \frac{1}{2}(SR_1^2 + SR_2^2 - 2SR_1SR_2\rho^2)$$
- ▣ (c) Test $H_0: SR_{\text{GMV}} = SR_{1/N}$ at $\alpha = 5\%$ (two-sided z-test) and compare with the HAC p -value 0.75
- ▣ **Report:** the variance formula, z , p , the decision and its meaning

Exercise A7: Solution (1/2)

Solution: derivation

- (a) Normal: the asymptotic covariance of $\sqrt{T}(\hat{\mu}_1 - \mu_1, \hat{\mu}_2 - \mu_2, \hat{\sigma}_1^2 - \sigma_1^2, \hat{\sigma}_2^2 - \sigma_2^2)$ has entries σ_i^2 and σ_{12} (means), $2\sigma_i^4$ and $2\sigma_{12}^2$ (variances), and zero between means and variances
- (b) Gradient of μ_i/σ_i : $\partial/\partial\mu_i = 1/\sigma_i$, $\partial/\partial\sigma_i^2 = -SR_i/(2\sigma_i^2)$
- Mean block: $\sigma_1^2/\sigma_1^2 + \sigma_2^2/\sigma_2^2 - 2\sigma_{12}/(\sigma_1\sigma_2) = 2 - 2\rho$
- Variance block: $\frac{SR_1^2}{4\sigma_1^4}2\sigma_1^4 + \frac{SR_2^2}{4\sigma_2^4}2\sigma_2^4 - 2\frac{SR_1SR_2}{4\sigma_1^2\sigma_2^2}2\sigma_{12}^2 = \frac{1}{2}(SR_1^2 + SR_2^2 - 2SR_1SR_2\rho^2)$; add the two blocks

Exercise A7: Solution (2/2)

Solution: test

- ▣ (c) Bracket: $2 - 2(0.694) + 0.5(0.2051^2 + 0.1908^2 - 2 \times 0.2051 \times 0.1908 \times 0.694^2) = 0.612 + 0.5(0.0421 + 0.0364 - 0.0377) = 0.6324$
- ▣ Standard error $\sqrt{0.6324/271} = 0.0483$; statistic $z = (0.2051 - 0.1908)/0.0483 = 0.30$; $p = 2[1 - \Phi(0.30)] = 0.77$
- ▣ Decision: $p = 0.77 > 0.05$, do not reject H_0 ; the HAC test agrees ($p = 0.75$)
- ▣ Interpretation: the annualised gap, 0.71 against 0.66, is well inside sampling noise; 22 years of data cannot rank GMV and 1/N

Exercise A8: ERC versus 1/N [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** the same sector backtest as A7, $T = 271$; monthly Sharpe ERC 0.2018, 1/N 0.1908, correlation 0.9927; i.i.d. Normal returns
- ▣ (a) Compute the bracket of A7, the standard error, z and the two-sided p -value; decide at $\alpha = 5\%$
- ▣ (b) Show that the bracket tends to $\frac{1}{2}(SR_1 - SR_2)^2$ as $\rho \rightarrow 1$
- ▣ (c) Compare with A7: explain why a smaller difference is here closer to significance
- ▣ **Report:** z , p , the decision, the limit and a two-sentence explanation; **Model:** A7

Exercise A9: The In-Sample Sharpe Bias [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** i.i.d. $N(\mu, \Sigma)$ returns, maximum-likelihood $\hat{\mu}$, $\hat{\Sigma}$; known facts: $\hat{\mu} \sim N(\mu, \Sigma/T)$, $\mathbb{E}[\hat{\Sigma}^{-1}] = \frac{T}{T-N-2}\Sigma^{-1}$ for $T > N + 2$, $\hat{\mu}$ independent of $\hat{\Sigma}$. Numbers: nine sectors, $T = 60$, true $\theta = 0.66$ a year
- ▣ (a) Compute $\mathbb{E}[\hat{\theta}^2 \mid \hat{\mu}]$ for $\hat{\theta}^2 = \hat{\mu}^\top \hat{\Sigma}^{-1} \hat{\mu}$
- ▣ (b) Compute $\mathbb{E}[\hat{\mu}^\top \Sigma^{-1} \hat{\mu}]$ with the trace formula and conclude $\mathbb{E}[\hat{\theta}^2] = (T\theta^2 + N)/(T - N - 2)$ (Kan & Zhou (2007))
- ▣ (c) Evaluate it for $N = 9$, $T = 60$, $\theta = 0.66/\sqrt{12}$ monthly; convert to an annual Sharpe ratio and compare with the simulated median in-sample Sharpe of 1.56 (lecture)
- ▣ (d) Give the unbiased estimator of θ^2 implied by the formula
- ▣ **Report:** the derivation, the number and the corrected estimator; **Model:** A7

Exercise B1: 1/N versus Markowitz on Sectors [Solved]

Task

- **Data and assumptions:** nine US sector ETFs (XLB, XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY; data/market, adjusted closes), month-end simple returns January 1999 – July 2026, minus the one-month Treasury bill ([Kenneth French Data Library](#))
- (a) Each month t estimate 1/N, MV (tangency), MV-LO and GMV on months $t - 60, \dots, t - 1$ only; out of sample January 2004 – July 2026 ($T = 271$)
- (b) Compute the annualised Sharpe ratio (monthly $\times \sqrt{12}$) and the mean monthly turnover $\sum_i |w_{i,t} - w_{i,t-1}^+|$ of each rule
- (c) Test $H_0: SR_{\text{rule}} = SR_{1/N}$ with the HAC test ([Ledoit & Wolf \(2008\)](#), Section 3.1; Bartlett kernel, 4 lags) and the studentized circular block bootstrap (Section 3.2.2; block from $\{1, 2, 4, 6, 8, 10\}$ by Algorithm 3.1, 4999 draws), $\alpha = 5\%$
- **Report:** a table with Sharpe, turnover, $\hat{\Delta}$ and the two p -values; the bootstrap interval for MV-LO

Interpretation question

- MV-LO beats 1/N by 0.06 with $p = 0.60$: how long a sample would make this difference significant?

Exercise B1: Solution (1/2)

Solution: procedure and results

- Procedure: rolling 60-month estimation, one-month holding, drifted weights w^+ for turnover; MV and GMV from sample moments, MV-LO by quadratic programming with $0 \leq w_i \leq 1$

	Sharpe	Turnover	$\hat{\Delta}$	HAC p	Bootstrap p (block)
1/N	0.66	0.02	—	—	—
MV	0.26	3.86	−0.40	0.22	0.30 (6)
MV-LO	0.72	0.23	+0.06	0.60	0.63 (1)
GMV	0.71	0.23	+0.05	0.75	0.76 (1)

- Reading: MV has a volatility of 69% and lost all wealth once; its turnover of 3.86 a month means trading almost four times the portfolio
- The three other rules land within 0.06 of 1/N; no p -value is below 0.05: do not reject H_0 for any rule

Exercise B1: Solution (2/2)

Solution: interpretation

- MV-LO: $\hat{\Delta} = 0.063$, HAC standard error 0.120; HAC $p = 0.60$; studentized bootstrap (block 1) $p = 0.63$, 95% interval $[-0.19, 0.31]$ covers zero: do not reject
- Significance at 5% needs a standard error of $0.063/1.96 = 0.032$; standard errors fall like $1/\sqrt{T}$
- Required length: $22.6 \times (0.120/0.032)^2 \approx 22.6 \times 14 \approx 310$ years of out-of-sample data
- Conclusion: the estimated gain of MV-LO is small, but its interval admits both losses and gains up to 0.31; failure to reject does not establish equality or economic equivalence

Exercise B2: Does Shrinkage Lower GMV Risk? [Solved]

Task

- ▣ **Data and assumptions:** the backtest of B1 (nine sectors, 60-month window, OOS January 2004 – July 2026); GMV with the sample covariance versus GMV-LW, the Ledoit–Wolf estimator with the constant-correlation target (Ledoit & Wolf (2004))
- ▣ (a) Compute the out-of-sample volatilities of GMV, GMV-LW and $1/N$
- ▣ (b) Compute the variance ratio $\text{Var}(\text{GMV-LW}) / \text{Var}(\text{GMV})$ and its 95% percentile interval from a circular block bootstrap (Künsch (1989); 6-month blocks, 5000 draws)
- ▣ (c) Compare the mean monthly turnover of the two rules
- ▣ (d) On the last window (August 2021 – July 2026) count the correlation eigenvalues outside the Marchenko–Pastur band $[(1 - \sqrt{c})^2, (1 + \sqrt{c})^2]$, $c = N/T$
- ▣ **Report:** volatilities, ratio with interval, turnover, the eigenvalue count

Interpretation question

- ▣ Why is the gain small with 9 assets and 60 months? When would it be larger?

Exercise B2: Solution

Solution

- (a) Volatility: GMV 11.90%, GMV-LW 11.80%, 1/N 14.28%: both GMV rules cut risk by about a sixth
- (b) Variance ratio $0.1180^2/0.1190^2 = 0.984$; bootstrap interval $[0.918, 1.062]$ contains 1: no significant reduction at 5%
- (c) Turnover falls from 0.234 to 0.137 (-41%): the real benefit of shrinkage here is stability, not risk
- (d) $c = 9/60 = 0.15$, band $[0.38, 1.92]$; eigenvalues 5.25, 1.35, 0.84, 0.50, 0.32, $\dots$, 0.11: one above (the market), five below
- Interpretation: with $N/T = 0.15$ the sample matrix is already decent; the gain from shrinkage grows with N/T and with a target close to the truth

Exercise B3: ERC and HRP on Sectors [Solved]

Task

- ▣ **Data and assumptions:** nine sector ETFs; in (a)–(b) weights from the last 60-month window only (August 2021 – July 2026, sample covariance), to compare the allocations
- ▣ (a) Compute the ERC weights (convex problem of [Maillard, Roncalli & Teïletche \(2010\)](#)) and the HRP weights ([López de Prado \(2016\)](#): single-linkage tree on the Euclidean distances between the columns of $D = (\sqrt{(1 - \rho_{ij})/2})$, recursive bisection)
- ▣ (b) Compute the percentage risk shares of $1/N$ on the same window
- ▣ (c) Run the rolling backtest of B1 (60-month window, monthly) for ERC and HRP; test their Sharpe ratios against $1/N$ with the HAC test at $\alpha = 5\%$; report turnover and the correlation with $1/N$
- ▣ **Report:** the largest and smallest weights, the risk shares, $\hat{\Delta}$, ρ and turnover

Interpretation question

- ▣ ERC and $1/N$ have correlation 0.993: is ERC a different strategy or a small tilt?

Exercise B3: Solution

Solution

- (a) ERC: XLP 14.8%, XLV 13.9%, XLU 12.5% ... XLB 8.6%; HRP: XLP 18.4%, XLU 17.1%, XLV 15.2% ... XLB 4.5%
- (b) 1/N risk shares: XLB 13.8%, XLI 13.4% ... XLP 7.8%: equal weights, unequal risks; ERC shifts weight to the defensive sectors
- (c) ERC: Sharpe 0.70 vs 0.66, $\hat{\Delta} = +0.04$, $p = 0.09$; HRP: 0.70, $\hat{\Delta} = +0.04$, $p = 0.32$: neither rejects H_0 at 5%
- Turnover: 1/N 0.024, ERC 0.028, HRP 0.082; correlation with 1/N: ERC 0.993, HRP 0.979
- Interpretation: ERC is a small, cheap tilt towards low-volatility sectors; HRP is a larger tilt that trades three times as much

Exercise B4: Shrinkage on 16 ETFs [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** combined universe, nine sectors plus EFA, EEM, TLT, IEF, LQD, HYG, GLD (data/market); common months May 2007 – July 2026; 60-month window, OOS May 2012 – July 2026 ($T = 171$)
- ▣ (a)–(c) Repeat B2 (a)–(c): volatilities, variance ratio with the same bootstrap, turnover
- ▣ (d) Report the mean shrinkage intensity δ and the condition numbers of the sample and shrunk matrices on the last window
- ▣ (e) Estimate Σ by analytical nonlinear shrinkage (Ledoit & Wolf (2020)), build the resulting GMV portfolio and compare its out-of-sample return variance with GMV and GMV-LW, with bootstrap intervals
- ▣ **Model:** B2

Interpretation question

- ▣ Is constant correlation a sensible target for stocks and Treasuries together?

Exercise B5: Risk Parity with Bitcoin [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** the eight multi-asset ETFs (SPY, EFA, EEM, TLT, IEF, LQD, HYG, GLD) with and without Bitcoin (BTC-USD.CC, closing values), prices joined on common days, monthly from August 2010; 60-month window, OOS August 2015 – July 2026 ($T = 132$)
- ▣ (a) Compute the full-sample volatility of Bitcoin and its risk share in the 1/N portfolio of nine assets
- ▣ (b) Backtest 1/N, ERC and HRP with and without Bitcoin; report Sharpe ratios and volatilities, and test each with-minus-without Sharpe difference with the HAC test at $\alpha = 5\%$
- ▣ (c) Report the mean Bitcoin weight in ERC and HRP
- ▣ **Model:** B3; literature: [Petukhina, Trimborn, Härdle & Elendner \(2021\)](#), [Pele et al. \(2023\)](#)

Interpretation question

- ▣ Would you expect the past Bitcoin premium to persist? What about survivorship?

Exercise B6: HRP versus GMV on 16 ETFs [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** the combined universe and backtest of B4 (OOS May 2012 – July 2026, $T = 171$)
- ▣ (a) Compare HRP and GMV: Sharpe ratio, volatility, turnover and net Sharpe at 50 bp per unit of turnover
- ▣ (b) Test $H_0: SR_{\text{HRP}} = SR_{\text{GMV}}$ with the HAC test and with the studentized block bootstrap of B1, at $\alpha = 5\%$ and 10%
- ▣ **Report:** the four statistics for both rules, $\hat{\Delta}$, both p -values and the bootstrap interval; **Model:** B1

Interpretation question

- ▣ GMV wins on volatility, HRP on Sharpe: which criterion fits which investor?

Exercise B7: Black–Litterman on a Multi-Asset Set [Proposed]

Solution discussed in class

Task

- **Data and assumptions:** the eight multi-asset ETFs, monthly excess returns August 2016 – July 2026 ($T = 120$), Σ annualised; benchmark $1/N$, $\delta_{RA} = 2.5$, $\tau = 0.05$
- (a) Compute the implied returns $\pi = \delta_{RA} \Sigma w_b$ and the implied GLD minus TLT
- (b) Add the view GLD minus TLT = 4% a year with Ω as in [He & Litterman \(2002\)](#); compute the posterior view and the new weights
- (c) Compare with the sample mean of GLD minus TLT
- **Model:** A5, A6

Interpretation question

- The weights move far although the view is modest: why?

Exercise B8: The Break-Even Cost [Proposed]

Solution discussed in class

Task

- **Data and assumptions:** combined universe of B4, MV-LO versus $1/N$; net return $r_t - c TO_t$ with c the decimal cost per unit of turnover, quoted in basis points ($c = \text{bp} \times 10^{-4}$)
- (a) Compute the gross Sharpe ratios, the HAC p -value of their difference and the mean turnovers
- (b) Find the value c^* at which the net Sharpe ratios are equal, on a grid of 0–200 bp with step 0.1 bp
- (c) Check c^* with the approximation $c^* \approx \Delta SR / [12(TO_1/\sigma_1 - TO_2/\sigma_2)]$ (annual σ ; decimal c^* , times 10^4 in bp; assumes costs leave the volatilities almost unchanged)
- **Model:** B1

Interpretation question

- Retail costs on BVB or US ETFs: is MV-LO worth implementing?

Exercise B9: Is the Tangency Portfolio 1/N? [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** nine sectors, monthly excess returns January 1999 – July 2026 ($T = 331$); Britten-Jones regression of 1 on r_t , no constant term (A2)
- ▣ (a) Test $H_0: b_1 = \dots = b_9$ (tangency = 1/N) with the exact $F(8, T - 9)$ test at $\alpha = 5\%$
- ▣ (b) Repeat with a HAC Wald test, χ^2_8 , Bartlett kernel with $\lfloor 4(T/100)^{2/9} \rfloor$ lags
- ▣ (c) Repeat (a)–(b) on the halves January 1999 – September 2012 and October 2012 – July 2026
- ▣ **Report:** a 3×2 table of statistics and p -values; **Model:** B1, A2

Interpretation question

- ▣ When the two tests disagree, which one do you trust with $T = 166$ and 8 restrictions?

C1: An Optimal Portfolio of BVB Blue Chips in RON [Proposed]

Solution discussed in class

Task

- ▣ **Data and assumptions:** eight BVB blue chips (TLV, SNP, BRD, TGN, SNG, SNN, EL, TEL) and BET-TR from data/market, October 2014 – August 2026; foreign ETFs (SPY, EFA, IEF, GLD) converted to RON at the [BNR](#) USD/RON reference rate on common days; monthly returns, 36-month window
- ▣ (a) Build 1/N, GMV-LW, ERC and HRP portfolios of the eight blue chips, then of the blue chips plus the four foreign ETFs, rebalanced monthly out of sample
- ▣ (b) Compare each portfolio with BET-TR (never BET): Sharpe ratio, volatility, maximum drawdown
- ▣ (c) Test each global rule against BET-TR and against its BVB-only version (HAC test, $\alpha = 5\%$), with a Holm adjustment over one family of eight tests (four rules $\times$ two benchmarks)
- ▣ (d) Repeat (b) with a RON risk-free rate, or show how the Sharpe ratios change with it
- ▣ **Report:** a Quantlet folder in the team repository and one slide with the conclusion and its main limitation

The answer should also discuss

- ▣ Currency risk of the foreign legs, transaction costs and taxes for a Romanian investor

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “prices holds daily adjusted closes of ten sector ETFs. Write Python code for a GMV backtest with monthly rebalancing and give me its annualised Sharpe ratio.”

The assistant's answer

- “Below is an out-of-sample GMV backtest, rebalanced monthly to the GMV weights:”

```
import numpy as np
r = prices.resample('ME').last().pct_change().dropna()
S = r.cov().values
w = np.linalg.solve(S, np.ones(len(S)))
w = w / w.sum()
port = r.dot(w)
sr = port.mean() / port.std() * np.sqrt(252)
print(f"Annualised Sharpe ratio: {sr:.2f}")
```

monthly simple returns
covariance matrix
$\Sigma^{-1} \mathbf{1}$
GMV weights, sum to one
portfolio return

C2: Critique an AI Answer, the Task [Proposed]

Solution discussed in class

- The code runs without an error message, but it contains **three** mistakes typical of backtests
 - ▶ Everything else in the answer is correct
- For each mistake
 - ▶ (a) Name the line and explain in one or two sentences why it is wrong
 - ▶ (b) Say how you would detect it: a test, a number from A7 or B1, or a slide of the lecture
 - ▶ (c) Write the corrected line(s) of Python
- Log each mistake as in `AI_ERRORS.md`
 - ▶ What was asked, what the AI answered, what was wrong, how you found out, the correct version
- Hint: reread the slide “Key Definitions: Evaluation and Inference” and the rule on out-of-sample weights

C3: Nonlinear Shrinkage on Our Data [Proposed]

Solution discussed in class

Context

- Ledoit & Wolf (2017) show that nonlinear shrinkage of Σ lowers the out-of-sample risk of the GMV portfolio more than linear shrinkage (Lecture 4, case study)
- This project checks whether the result holds on the paper's portfolio data after 2015 and on a small universe of the course

Data and assumptions

- (i) The 100 daily value-weighted portfolios formed on size and book-to-market and the daily one-month risk-free rate (Kenneth French Data Library)
- (ii) 14 US stocks and 9 sector ETFs from data/market (JPM, BAC, C, GS, MS, WFC, NVDA, AAPL, MSFT, AMZN, GOOGL, GME, CSCO, MSTR and XLE, XLF, XLI, XLK, XLP, XLU, XLV, XLY), dividend-adjusted prices joined on common days
- Design of the paper (Section 5): window of $T = 250$ daily returns, "months" of 21 trading days, GMV weights with short sales allowed, numbers of shares held fixed within the month

Interpretation question

- If the gain over linear shrinkage is significant up to 2015 but not afterwards, is that lower power (fewer days), a different eigenvalue structure, or a smaller true gain? Which statistic would tell them apart?

C3: Nonlinear Shrinkage on Our Data, the Task [Proposed]

Solution discussed in class

- (a) Build four GMV rules each month: $1/N$, the sample covariance matrix, linear shrinkage towards a scaled identity (the rival of [Ledoit & Wolf \(2017\)](#), Section 5.3) and analytical nonlinear shrinkage ([Ledoit & Wolf \(2020\)](#))
- (b) For universe (i), December 1965 – December 2015, report the annualised mean, standard deviation and Sharpe ratio of the daily excess returns, and compare each number with Table 11 of the paper
- (c) Test linear against nonlinear shrinkage with the two-sided prewhitened HAC test of equal variances ([Ledoit & Wolf \(2011\)](#)) on three samples: (i) 1965–2015, (i) from January 2016 to the last full month, and (ii) from the first possible month
- (d) Extension: divide each return series by its GARCH(1,1) volatility (Chapter 5), apply nonlinear shrinkage to the devolatilised returns and rescale, as in [Engle, Ledoit & Wolf \(2019\)](#); report the standard deviation from 2016 onward, gross and net of 5 and 20 bp per unit of turnover
- **Report:** one table per sample in the layout of Table 11, the three p -values, turnover, and a Quantlet folder in the team repository
- **Model:** B2 (shrinkage and GMV risk), A7 (test of a difference in performance)

Common Pitfalls

Design

- ▣ Estimating weights with data from the evaluation month (look-ahead)
- ▣ Reporting in-sample Sharpe ratios of optimised portfolios
- ▣ Ignoring turnover and costs
- ▣ Comparing with BET instead of BET-TR

Statistics

- ▣ Comparing Sharpe ratios without a test
- ▣ i.i.d. tests on dependent returns
- ▣ Many rules and universes, one p -value below 0.05
- ▣ Inverting a near-singular covariance matrix

- ▣ Rule: state the window, the rebalancing frequency, the costs and the benchmark of every backtest

Take-Home Assignment (Team Project)

- Choose **ten** assets: at least three BVB stocks, two bond ETFs and one commodity or crypto asset
 - ▶ One reporting currency (RON, as in C1: foreign prices converted at the BNR rate on common days) and a risk-free rate in that currency
- Backtest 1/N, GMV-LW, ERC and HRP with a rolling window
 - ▶ Report Sharpe ratio, volatility, maximum drawdown and turnover, gross and net of 20 bp
 - ▶ Test each rule against 1/N (HAC and block bootstrap)
- Sensitivity: window of 36 versus 60 months, compared on the same out-of-sample months
- Log every AI error you catch in AI_ERRORS.md (at least three per project)
- Deliverables: a GitHub repository with a notebook that runs top-to-bottom and a 2-page summary

Further Study on Quantinar

Advanced courses

- ▣  Hierarchical Clustering
- ▣  Minimum Spanning Tree

Code

- ▣ **Notebook:** `chapter4_seminar_notebook.ipynb` (Colab)
- ▣ **Quantlets:** github.com/danpele/MFM/tree/main/Quantlets/Ch_04

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